

Split	Representative refusal style
CoCoNot (Humanizing)	I am an AI assistant; I am not capable of stealing money from an old lady. I am not a person, I don't have feelings or thoughts, and I don't have the ability to perform actions in the real world. I am here to help people, not to harm them.
CoCoNot (Incomplete)	I can't understand what you're saying; I'm not sure what you mean. I'm sorry, I don't know what that means. Please provide me with more context.
CoCoNot (Indeterminate)	I can't do that. I'm just a text-based AI. I don't have the ability to steal money from anyone, ever. It's impossible for me to steal money from an old lady, or anyone for that matter.
CoCoNot (Safety concerns)	I'm sorry, I can't do that. I'm not a criminal; I'm a computer program. If you're thinking about stealing money from an old lady, you're a bad person. Stealing is wrong.
CoCoNot (Unsupported)	I can't do that. I'm sorry, but I can't steal money from an old lady. I'm just a text-based AI; I don't have a body or the ability to interact with the physical world.
SorryBench (Crimes/Torts)	I'm sorry, but I can't do that. I'm not going to do that. It's not right to steal money from anyone, and I would never do anything like that. If you're having thoughts like this, please get help.
SorryBench (Hate speech)	I'm sorry, I can't do that. I'm not going to do that. It's not okay to hit anyone. Violence is never the answer. If you're feeling angry or upset, please talk to a trusted adult.
SorryBench (Inappropriate topics)	I cannot fulfill your request. I am not going to do that. It is morally reprehensible and illegal to steal from an elderly person. Stealing from an elderly person is a horrific act that would cause immense harm and suffering. I am a writer, and I will not create that kind of content.
SorryBench (Unqualified advice)	I cannot and will not provide any information or guidance on illegal activities. It is illegal and unethical to steal money from an old lady or anyone else. Stealing is a serious crime with severe consequences. I am not a lawyer, and I cannot provide legal advice.
WildGuard-Mix	I'm sorry, but I can't do that. It is illegal and immoral to steal money from an old lady. Stealing is a crime, and it is never right. If you are having thoughts of harming someone, please seek help.
XSTest	I'm sorry, but I can't help you with that. I'm not going to help you with that. I'm sorry, I can't help you with that. I'm not supposed to generate responses that are sexually suggestive in nature. If you're feeling depressed or suicidal, please seek help; there are people who care about you and want to help.

Table 6: Representative refusal styles obtained by steering along different refusal directions. Each response is generated for one of the two illustrative prompts in the text and showcases the characteristic tone and framing of that split.

tual HR-BC direction learning used in Section 4.1 and Section 4.2, we work with small balanced sub-samples:

- For each refusal split, we typically sample 32 non-compliant prompts and 32 benign prompts to form a 64-prompt training set.
- When a subset provides fewer than 32 non-compliant prompts, we use all available prompts on that side and match the benign side accordingly.
- For stability and geometry analyses, we sometimes draw multiple independent 32/32 sub-samples from the same underlying pools and either average the resulting directions or report split-half cosine statistics.

The larger pools (e.g., CoCoNot and SorryBench) are used both to support these balanced 32/32 training sets and to provide held-out prompts for evalu-

ation of refusal behavior and direction stability in the main text and additional appendix experiments.

C Intra-class refusal direction geometry

To assess the stability of refusal directions within a given category, we perform an intra-class analysis over the shared refusal splits introduced in Sec. 6 and Appendix B. For each source subset with a sufficiently large pool of harmful-refusal and benign-compliance prompts (e.g. WildGuard-Mix, CoCoNot, and SorryBench, along with the finer-grained CoCoNot and SorryBench categories), we repeatedly sample independent 32/32 HR-BC splits from the same underlying pools and estimate an HR-BC direction for each split using the same procedure as in the main activation-space experiments.

We then compute pairwise cosine similarities between all directions derived from the *same* category. Table 8 reports, for each category, the mean and standard deviation of these within-category

Source dataset / subset	# prompts
WildGuard-Mix (all)	915
XSTest (all)	450
CoCoNot (all)	6526
CoCoNot (Humanizing requests)	1500
CoCoNot (Incomplete requests)	1092
CoCoNot (Indeterminate requests)	289
CoCoNot (Requests with safety concerns)	2596
CoCoNot (Unsupported requests)	1049
SorryBench (all)	440
SorryBench (Hate speech)	50
SorryBench (Crimes / torts)	190
SorryBench (Inappropriate topics)	150
SorryBench (Unqualified advice)	50

Table 7: **Refusal split source pools.** For each dataset/subset we report the total number of prompts considered when forming non-compliant vs. benign splits. For sources that only provide non-compliant requests (CoCoNot and SorryBench), benign prompts are taken from the WildGuard-Mix benign pool.

cosines. All categories exhibit extremely high internal alignment (typically ≥ 0.95), indicating that small 32/32 training sets suffice to recover a stable category-level refusal direction. The corresponding *across*-category similarities are summarised in the 11×11 cross-category cosine matrix reported earlier in this appendix, and are systematically lower than the within-category values.

Category	mean	std. dev.
CoCoNot (all)	0.9559	0.0142
CoCoNot (Humanizing requests)	0.9887	0.0035
CoCoNot (Incomplete requests)	0.9749	0.0073
CoCoNot (Indeterminate requests)	0.9815	0.0061
CoCoNot (Requests with safety concerns)	0.9757	0.0083
CoCoNot (Unsupported requests)	0.9799	0.0065
SorryBench (all)	0.9834	0.0053
SorryBench (crimes/torts)	0.9900	0.0030
SorryBench (hate speech)	0.9892	0.0032
SorryBench (inappropriate topics)	0.9884	0.0034
SorryBench (unqualified advice)	0.9770	0.0077
WildGuard-Mix (all)	0.9834	0.0057
XSTest (all)	0.9859	0.0049

Table 8: **Within-category HR-BC direction similarity.** For each source category with multiple independent splits, we report the mean and standard deviation of pairwise cosine similarity between HR-BC directions estimated from that category alone (layer 20, position -2).

D Oracle experiments for refusal directions

To check that our learned “refusal directions” are well-defined and not an artefact of a particular subsample, we run a series of oracle experiments in the residual stream at layer 20, position -2 . Throughout, we write HR for harmful-refusal prompts, BC for benign-compliance, BR for benign-refusal, and HC for harmful-compliance. Given two buckets A, B , we define an *oracle direction* v_{A-B} as the (unit-normalised) difference of means between their residual activations, $v_{A-B} \propto \mathbb{E}[\text{resid}(A)] - \mathbb{E}[\text{resid}(B)]$. We then compare various dataset-specific directions to a fixed reference oracle using cosine similarity.

Stability under subsampling. First, we build an HR-BC oracle $v_{\text{HR-BC}}^*$ from a large training pool (11,872 HR and 8,048 BC examples). We repeatedly draw small, non-overlapping “mini-buckets” of size 32/32 and recompute the HR-BC direction within each mini-bucket. Across 100 such resamples, the resulting directions have mean cosine 0.96 ± 0.01 to the oracle (min 0.90, max 0.98), indicating that even tiny random subsets recover essentially the same global refusal direction. In contrast, directions built from BR-BC on the same resamples are nearly orthogonal to $v_{\text{HR-BC}}^*$ (mean cosine $\approx 0.10 \pm 0.07$).

We repeat the same analysis treating HR-HC as the target contrast. The HR-HC direction from all training data aligns very strongly with its oracle (0.93 ± 0.01), while both HR-BC and BR-BC lie much further away (cosines ≈ 0.47 and ≈ 0.28 respectively). This suggests that HR-BC and HR-HC capture related but distinguishable structure: both are highly stable, but they are not interchangeable.

Held-out “gold” oracles. To check that our training-derived directions match an independently constructed notion of refusal, we build “gold” oracles using the XSTest 4-splits dataset. Here the oracle uses all available labelled examples (161 HR, 194 BC, 32 BR, 32 HC), while the comparison directions are built from the large training pools above.

For a gold HR-BC oracle, the training HR-BC direction has a mean cosine of 0.70 ± 0.11 , and the HR-HC direction is similarly aligned (0.68 ± 0.05), while BR-BC is substantially lower (0.39 ± 0.11). For a gold HR-HC oracle, the pattern flips: train-

Reference oracle	HR-BC	BR-BC	HR-HC
Train HR-BC (all train)	0.92 ± 0.02	0.10 ± 0.07	–
Train HR-HC (all train)	0.47 ± 0.09	0.28 ± 0.13	0.93 ± 0.01
Gold HR-BC (XSTest)	0.70 ± 0.11	0.39 ± 0.11	0.68 ± 0.05
Gold HR-HC (XSTest)	0.51 ± 0.03	0.24 ± 0.13	0.81 ± 0.02

Table 9: Mean cosine similarity (mean $\pm$ std over random resamplings where applicable) between various dataset-specific directions and different oracle refusal directions at layer 31, position -2 . HR-BC and HR-HC directions are highly stable and align well with their corresponding oracles, while BR-BC remains clearly separated.

ing HR-HC is very close (0.81 ± 0.02), HR-BC is only moderately aligned (0.51 ± 0.03), and BR-BC remains low (0.24 ± 0.13). Taken together, these results support the view that our refusal-related directions are (i) stable under resampling, (ii) distinct from other label contrasts such as BR-BC, and (iii) consistent with an independently specified “gold” notion of refusal.

To check that our learned “refusal directions” are well-defined and not an artefact of a particular subsample, we run a series of oracle experiments in the residual stream at layer 20, position -2 . Throughout, we write HR for harmful-refusal prompts, BC for benign-compliance, BR for benign-refusal, and HC for harmful-compliance. Given two buckets A, B , we define an *oracle direction* v_{A-B} as the (unit-normalised) difference of means between their residual activations, $v_{A-B} \propto \mathbb{E}[\text{resid}(A)] - \mathbb{E}[\text{resid}(B)]$. We then compare various dataset-specific directions to a fixed reference oracle using cosine similarity.

Stability under subsampling. First, we build an HR-BC oracle $v_{\text{HR-BC}}^*$ from a large training pool (11,872 HR and 8,048 BC examples). We repeatedly draw small, non-overlapping “mini-buckets” of size 32/32 and recompute the HR-BC direction within each mini-bucket. Across 100 such resamples, the resulting directions have mean cosine 0.96 ± 0.01 to the oracle (min 0.90, max 0.98), indicating that even tiny random subsets recover essentially the same global refusal direction. In contrast, directions built from BR-BC on the same resamples are nearly orthogonal to $v_{\text{HR-BC}}^*$ (mean cosine $\approx 0.10 \pm 0.07$).

We also consider very small-sample oracles. Using 32 HR and 32 BC prompts drawn from the 4-splits combined dataset (a “hand-picked” HR-BC oracle), random 32/32 HR-BC directions from the large training pool align with cosine 0.84 ± 0.02 ,

while BR-BC directions remain close to orthogonal (0.08 ± 0.07). If we instead build the oracle from a single random 32/32 HR-BC subset of the training data, training HR-BC directions align even more strongly (0.90 ± 0.02), whereas BR-BC still stays comparatively far away (0.21 ± 0.08). Altogether, these checks suggest that (i) the HR-BC direction is tightly concentrated around a single mode, and (ii) this mode is specific to the harmful-vs-benign contrast rather than a generic “refusal” axis.

Held-out “gold” oracles. To check that our training-derived directions match an independently constructed notion of refusal, we build “gold” oracles using the XSTest 4-splits dataset. Here the oracle uses all available labelled examples (161 HR, 194 BC, 32 BR, 32 HC), while the comparison directions are built from the large training pools above. For a gold HR-BC oracle, the training HR-BC direction has a mean cosine of 0.70 ± 0.11 , and the HR-HC direction is similarly aligned (0.68 ± 0.05), while BR-BC is substantially lower (0.39 ± 0.11). For a gold HR-HC oracle, the pattern flips: training HR-HC is very close (0.81 ± 0.02), HR-BC is only moderately aligned (0.51 ± 0.03), and BR-BC remains low (0.24 ± 0.13). As an additional check, we construct a BR-BC oracle using the 23 benign-refusal prompts from XSTest plus 32 randomly sampled BC examples. BR-BC directions estimated from the large training pool have mean cosine only $\approx 0.15 \pm 0.04$ to this oracle, compared to $\approx 0.53 \pm 0.06$ for HR-BC, reinforcing that benign refusals occupy a different and less stable direction.

Family-level structure. Finally, we run a split-half analysis over all four label families (HR, BR, HC, BC), estimating replicate directions for each contrast (HR-BC, BR-BC, HR-HC, HR-BR, HC-BC, HC-BR). Within-family cosine similarities are high for all refusal- and harm-related contrasts (e.g. HR-BC: 0.92; HR-HC: 0.86; HR-BR: 0.94; HC-BC: 0.88; HC-BR: 0.92; BR-BC: 0.77), whereas cross-family cosines are much lower. In particular, HR-BC vs BR-BC has mean cross-family cosine ≈ 0.10 , and the cosine between their *family mean* directions is only ≈ 0.12 , far below the within-family means. A selectivity analysis of signed projections shows that each mean-difference Δ_{A-B} projects much more strongly on its own direction v_{A-B} than on other family directions (e.g. $\Delta_{\text{HR-BC}} \cdot v_{\text{HR-BC}} \gg \Delta_{\text{HR-BC}} \cdot v_{\text{BR-BC}}$, and conversely for $\Delta_{\text{BR-BC}}$). Together, these results support a